

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

Criteria	Understanding of Case (2.5)	Application of Concepts (2.5)	Analysis (2)	Solution Design (2)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						

Code of Conduct:

*I S. Blewke (19252525) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: S. Blewke.

CO3 - Assessment - 2

Pre Act	Y	N
Y	TP Real wolf came, warning was also given	FN wolf came, warning was not acted upon
N	FP No wolf, false warnings given	TN wolf didn't come warning was not given

2A) construct the matrix,

	Y	N
Y	TP 25	FN 20
N	FP 5	TN 50

→ Actual cat images = 30

→ Total = 100

→ Non-cat images = $100 - 30 = 70$

i) precision: $\frac{TP}{TP+FP} = \frac{25}{25+5} = \frac{25}{30} = 0.83$

(ii) Recall: $\frac{TP}{TP+FN} = \frac{25}{25+20} = \frac{25}{45} = 0.55$

iii) F_{score} : $2 \times \frac{\text{Precision} \times \text{recall}}{\text{Precision} + \text{recall}}$

$$= \frac{2 \times 0.83 \times 0.55}{0.83 + 0.55} = \frac{0.913}{1.38} = 0.66$$

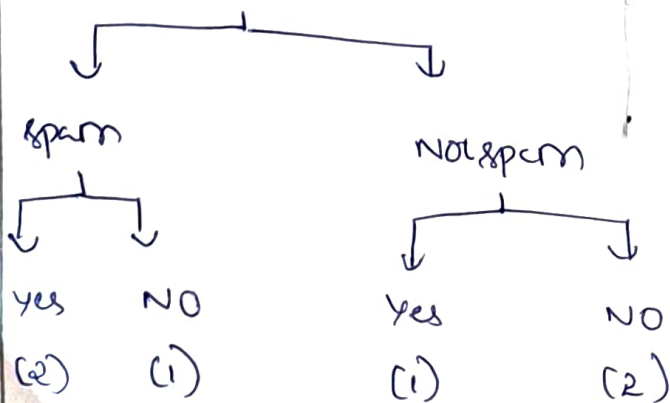
3) i) prior probability:-

$$P(\text{spam}) = \frac{3}{6}$$

$$P(\text{not spam}) = \frac{3}{6}$$

ii) compute conditional probabilities for each feature

i) offer:



Probability:-

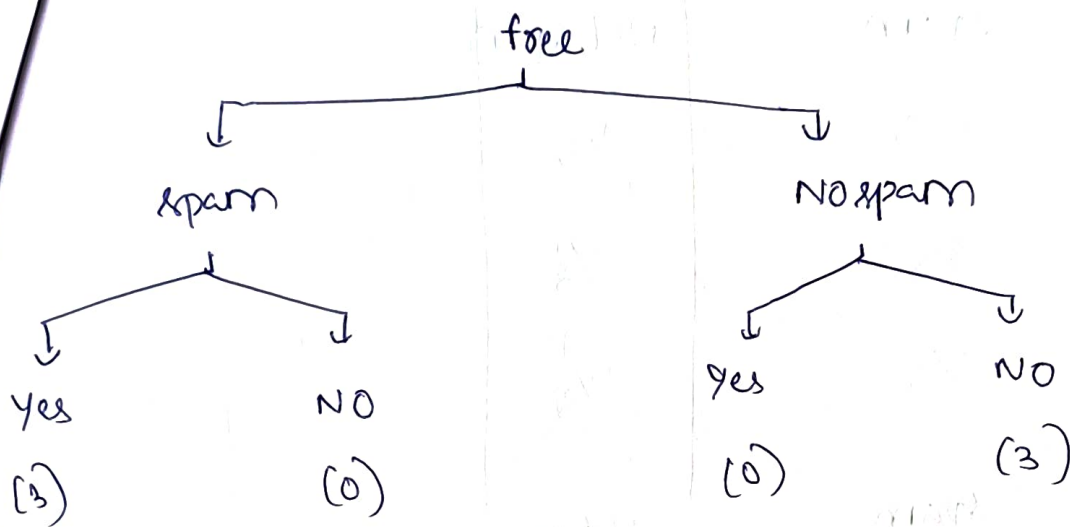
$$P(\text{yes} / \text{spam}) = \frac{2}{3}$$

$$P(\text{yes} / \text{not spam}) = \frac{1}{3}$$

$$P(\text{no} / \text{spam}) = \frac{1}{3}$$

$$P(\text{no} / \text{not spam}) = \frac{2}{3}$$

conditional probability for feature free:-



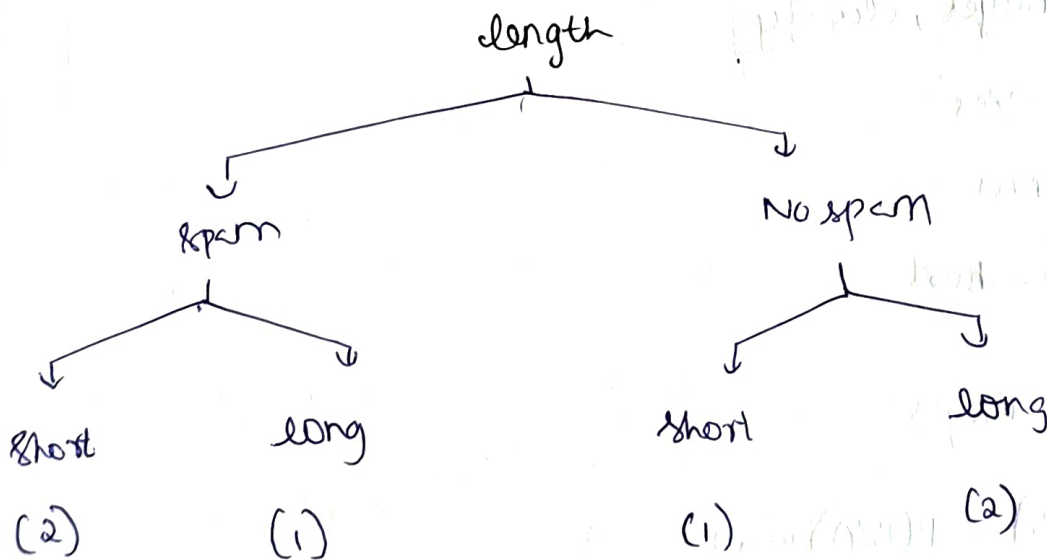
$$P(\text{Yes}/\text{span}) = 3/3$$

$$P(\text{Yes}/\text{No span}) = 0/3$$

$$P(\text{No}/\text{span}) = 0/3$$

$$P(\text{No}/\text{No span}) = 3/3$$

iii) conditional probability for feature length



$$P(\text{Short}/\text{span}) = 2/3$$

$$P(\text{Short}/\text{No span}) = 1/3$$

$$P(\text{Long}/\text{span}) = 1/3$$

$$P(\text{Long}/\text{No span}) = 2/3$$

Dataset

(1) Bayes.Na

(2) Bayes

iris - dataset

(100) 95.53

95.47

(V/L)

(0/1/0)

key:-

1) Bayes Naive Bayes "599523120178567155"

2) Bayes Naive Bayes Multinomial "5932177440181257085"

